Complete Maven Mastery

1. Java Project Build Using Maven Archetype

```
Command Prompt

D:\mavenprojects>mvn archetype:generate -DgroupId=com.example -DartifactId=my-app1 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: D:\mavenprojects
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: my-app1
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: D:\mavenprojects\my-app1
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.387 s
[INFO] Finished at: 2026-03-25T17:12:51+05:30
[INFO] ------------------------------------------------------------------------
                d
'd' is not recognized as an internal or external command,
operable program or batch file.

D:\mavenprojects>dir
 Volume in drive D is New Volume
 Volume Serial Number is AEE6-4A19

 Directory of D:\mavenprojects

25-03-2026  17:12    <DIR>          .
25-03-2026  17:12    <DIR>          my-app1
               0 File(s)              0 bytes
               2 Dir(s)  207,649,878,016 bytes free

D:\mavenprojects>cd my-app1
```

```
Command Prompt

25-03-2026  17:12    <DIR>          my-app1
               0 File(s)              0 bytes
               2 Dir(s)  207,649,878,016 bytes free

D:\mavenprojects>cd my-app1

D:\mavenprojects\my-app1>mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< com.example:my-app1 >-------------------------
[INFO] Building my-app1 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.4/maven-install-plugin-3.1.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.4/maven-install-plugin-3.1.4.pom (8.1 kB at
5.7 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/43/maven-plugins-43.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/43/maven-plugins-43.pom (7.5 kB at 63 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.4/maven-install-plugin-3.1.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.4/maven-install-plugin-3.1.4.jar (33 kB at 8
9 kB/s)
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ my-app1 ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ my-app1 ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\mavenprojects\my-app1\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ my-app1 ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ my-app1 ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\mavenprojects\my-app1\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ my-app1 ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ my-app1 ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit.JUnit3Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
```

```
Command Prompt
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.015 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ my-app1 ---
[INFO] Building jar: D:\mavenprojects\my-app1\target\my-app1-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- install:3.1.4:install (default-install) @ my-app1 ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.pom (2.2 kB at
 7.1 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.22/maven-resolver-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.22/maven-resolver-1.9.22.pom (23 kB at 115 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.pom (2.2 kB at 1
7 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar (196 kB at
 595 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar (157 kB at 8
09 kB/s)
[INFO] Installing D:\mavenprojects\my-app1\pom.xml to C:\Users\sreev\.m2\repository\com\example\my-app1\1.0-SNAPSHOT\my-app1-1.0-SNAPSHOT.pom
[INFO] Installing D:\mavenprojects\my-app1\target\my-app1-1.0-SNAPSHOT.jar to C:\Users\sreev\.m2\repository\com\example\my-app1\1.0-SNAPSHOT\my-app1-1.0-SNA
PSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  10.475 s
[INFO] Finished at: 2026-03-25T17:27:53+05:30
[INFO] ------------------------------------------------------------------------

D:\mavenprojects\my-app1>java -cp target/my-app1-1.0-SNAPSHOT.jar com.example.App
Hello World!

D:\mavenprojects\my-app1>
```

2. Maven-Docker Integration for Containerized Java Application

```
Command Prompt
D:\mavenprojects>mvn archetype:generate -DgroupId=com.example -DartifactId=docker-demo -DarchetypeArtifactId=maven-arche
type-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: D:\mavenprojects
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: docker-demo
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: D:\mavenprojects\docker-demo
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.147 s
[INFO] Finished at: 2026-03-26T16:58:45+05:30
```

```
Command Prompt

D:\mavenprojects\docker-demo>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:docker-demo >-----------------------
[INFO] Building docker-demo 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ docker-demo ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ docker-demo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\mavenprojects\docker-demo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ docker-demo ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ docker-demo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\mavenprojects\docker-demo\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ docker-demo ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO]
```

```
Command Prompt

 Volume in drive D is New Volume
 Volume Serial Number is AEE6-4A19

 Directory of D:\mavenprojects\docker-demo\src\main\java\com\example

26-03-2026  16:58    <DIR>          .
26-03-2026  16:58    <DIR>          ..
26-03-2026  16:58               187 App.java
               1 File(s)            187 bytes
               2 Dir(s)  207,647,830,016 bytes free

D:\mavenprojects\docker-demo\src\main\java\com\example>notepad App.java

D:\mavenprojects\docker-demo\src\main\java\com\example>cd ..

D:\mavenprojects\docker-demo\src\main\java\com>cd ..

D:\mavenprojects\docker-demo\src\main\java>cd ..

D:\mavenprojects\docker-demo\src\main>cd ..

D:\mavenprojects\docker-demo\src>cd ..

D:\mavenprojects\docker-demo>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:docker-demo >-----------------------
[INFO] Building docker-demo 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
```

```
Command Prompt
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.032 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ docker-demo ---
[INFO] Building jar: D:\mavenprojects\docker-demo\target\docker-demo-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.539 s
[INFO] Finished at: 2026-03-26T17:07:24+05:30
[INFO] ------------------------------------------------------------------------

D:\mavenprojects\docker-demo>dir
 Volume in drive D is New Volume
 Volume Serial Number is AEE6-4A19

 Directory of D:\mavenprojects\docker-demo

26-03-2026  17:07    <DIR>          .
26-03-2026  16:58    <DIR>          ..
26-03-2026  16:58               664 pom.xml
26-03-2026  16:58    <DIR>          src
26-03-2026  17:07    <DIR>          target
               1 File(s)            664 bytes
```

```
Command Prompt
[INFO]
[INFO]  ---> 609aa7e46fe1
[INFO] Step 4/4 : ENTRYPOINT ["java","-jar","app.jar"]
[INFO]
[INFO]  ---> Running in 8205761256b3
[INFO]  ---> Removed intermediate container 8205761256b3
[INFO]  ---> 0502589a4151
[INFO] Successfully built 0502589a4151
[INFO] Successfully tagged docker-demo:1.0-SNAPSHOT
[INFO]
[INFO] Detected build of image with id 0502589a4151
[INFO] Building jar: D:\mavenprojects\docker-demo\target\docker-demo-1.0-SNAPSHOT-docker-info.jar
[INFO] Successfully built docker-demo:1.0-SNAPSHOT
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  32.102 s
[INFO] Finished at: 2026-03-26T19:06:05+05:30
[INFO] ------------------------------------------------------------------------

D:\mavenprojects\docker-demo>docker run docker-demo:1.0-SNAPSHOT
Docker-Maven integration
U are viewing magic of cse
Step: 0
Step: 1
Step: 2
Step: 3
Step: 4

D:\mavenprojects\docker-demo>
```

Command Prompt

```
D:\mavenprojects\docker-demo>type dockerfile
FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY target/docker-demo-1.0-SNAPSHOT.jar app.jar
ENTRYPOINT ["java","-jar","app.jar"]
D:\mavenprojects\docker-demo>type pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/maven-v4_0_0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>docker-demo</artifactId>
    <version>1.0-SNAPSHOT</version>
    <packaging>jar</packaging>

    <name>docker-demo</name>
    <url>http://maven.apache.org</url>

    <dependencies>
        <!-- JUnit for testing -->
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>3.8.1</version>
            <scope>test</scope>
        </dependency>
```

Command Prompt

```
    <build>
        <plugins>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.3.0</version>
                <configuration>
                    <archive>
                        <manifest>
                            <mainClass>com.example.App</mainClass>
                        </manifest>
                    </archive>
                </configuration>
            </plugin>

            <!-- Dockerfile Maven Plugin -->
            <plugin>
                <groupId>com.spotify</groupId>
                <artifactId>dockerfile-maven-plugin</artifactId>
                <version>1.4.13</version>

                <configuration>
                    <repository>docker-demo</repository>
                    <tag>${project.version}</tag>
                </configuration>

                <executions>
                    <execution>
                        <id>default</id>
```

```
Command Prompt
                </configuration>
            </plugin>

            <!-- Dockerfile Maven Plugin -->
            <plugin>
                <groupId>com.spotify</groupId>
                <artifactId>dockerfile-maven-plugin</artifactId>
                <version>1.4.13</version>

                <configuration>
                    <repository>docker-demo</repository>
                    <tag>${project.version}</tag>
                </configuration>

                <executions>
                    <execution>
                        <id>default</id>
                        <goals>
                            <goal>build</goal>
                        </goals>
                    </execution>
                </executions>

            </plugin>

        </plugins>
    </build>

</project>
D:\mavenprojects\docker-demo>
```

App.java modification and image building for practice:

```
Command Prompt
[INFO] Step 2/4 : WORKDIR /app
[INFO]
[INFO]  ---> Using cache
[INFO]  ---> ff885608776b
[INFO] Step 3/4 : COPY target/docker-demo-1.0-SNAPSHOT.jar app.jar
[INFO]
[INFO]  ---> 72401a51a13a
[INFO] Step 4/4 : ENTRYPOINT ["java","-jar","app.jar"]
[INFO]
[INFO]  ---> Running in 96ee1d95767d
[INFO]  ---> Removed intermediate container 96ee1d95767d
[INFO]  ---> 6204b3f4063c
[INFO] Successfully built 6204b3f4063c
[INFO] Successfully tagged docker-demo:1.0-SNAPSHOT
[INFO]
[INFO] Detected build of image with id 6204b3f4063c
[INFO] Building jar: D:\mavenprojects\docker-demo\target\docker-demo-1.0-SNAPSHOT-docker-info.jar
[INFO] Successfully built docker-demo:1.0-SNAPSHOT
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  38.254 s
[INFO] Finished at: 2026-03-27T12:38:50+05:30
[INFO] ------------------------------------------------------------------------

D:\mavenprojects\docker-demo>docker run docker-demo:1.0-SNAPSHOT
Docker-Maven integration
U are viewing magic of cse
It is a palindrome
```

3. Containerization and Deployment of a Java Web Application using Docker and Apache Tomcat

```
Command Prompt

C:\Users\sreev>mvn archetype:generate -DgroupId=com.example -DartifactId=javawebapp -DarchetypeArtifactId=maven-archetyp
e-webapp -DarchetypeVersion=1.4 -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Archetype: maven-archetype-webapp:1.4
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: javawebapp
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: packageInPathFormat, Value: com/example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: javawebapp
[INFO] Project created from Archetype in dir: C:\Users\sreev\javawebapp
[INFO] ----------------------------------------------------------------------------
```

```
Command Prompt

C:\Users\sreev>cd javawebapp

C:\Users\sreev\javawebapp>dir
 Volume in drive C is OS
 Volume Serial Number is 0CE8-3131

 Directory of C:\Users\sreev\javawebapp

31-03-2026  14:06    <DIR>          .
31-03-2026  14:06    <DIR>          ..
31-03-2026  14:06             2,271 pom.xml
31-03-2026  14:06    <DIR>          src
               1 File(s)          2,271 bytes
               3 Dir(s)  44,589,912,064 bytes free

C:\Users\sreev\javawebapp>cd src/main/webapp

C:\Users\sreev\javawebapp\src\main\webapp>notepad index.jsp

C:\Users\sreev\javawebapp\src\main\webapp>cd ..

C:\Users\sreev\javawebapp\src\main>cd ..

C:\Users\sreev\javawebapp\src>cd /

C:\>cd \Users\sreev\javawebapp

C:\Users\sreev\javawebapp>notepad pom.xml

C:\Users\sreev\javawebapp>dir
```

```
Command Prompt

C:\Users\sreev\javawebapp>notepad dockerfile

C:\Users\sreev\javawebapp>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.safera.app:javawebapp >----------------------
[INFO] Building Java Web Application 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ war ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.8.1/mave
n-compiler-plugin-3.8.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.8.1/maven
-compiler-plugin-3.8.1.pom (12 kB at 19 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/33/maven-plugins-3
3.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/33/maven-plugins-33
.pom (11 kB at 227 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.8.1/mave
n-compiler-plugin-3.8.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.8.1/maven
-compiler-plugin-3.8.1.jar (62 kB at 914 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.3.2/maven-war
-plugin-3.3.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.3.2/maven-war-
plugin-3.3.2.pom (9.7 kB at 242 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/34/maven-plugins-3
4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/34/maven-plugins-34
.pom (11 kB at 243 kB/s)
```

```
Command Prompt
jar (263 kB at 474 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/sisu/org.eclipse.sisu.plexus/0.0.0.M2a/org.ecl
ipse.sisu.plexus-0.0.0.M2a.jar (202 kB at 365 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/sisu/sisu-guice/3.1.0/sisu-guice-3.1.0-no_aop
.jar (357 kB at 644 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/asm/asm/3.3.1/asm-3.3.1.jar (44 kB at 79 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/sisu/org.eclipse.sisu.inject/0.0.0.M2a/org.ecl
ipse.sisu.inject-0.0.0.M2a.jar (202 kB at 364 kB/s)
[INFO] Packaging webapp
[INFO] Assembling webapp [javawebapp] in [C:\Users\sreev\javawebapp\target\javawebapp]
[INFO] Processing war project
[INFO] Copying webapp resources [C:\Users\sreev\javawebapp\src\main\webapp]
[INFO] Building war: C:\Users\sreev\javawebapp\target\javawebapp.war
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  9.037 s
[INFO] Finished at: 2026-03-31T14:14:43+05:30
[INFO] ------------------------------------------------------------------------

C:\Users\sreev\javawebapp>dir
 Volume in drive C is OS
 Volume Serial Number is 0CE8-3131

 Directory of C:\Users\sreev\javawebapp

31-03-2026  14:14    <DIR>          .
31-03-2026  14:06    <DIR>          ..
31-03-2026  14:14               156 dockerfile.txt
31-03-2026  14:09             2,407 pom.xml
```

```
Command Prompt

C:\Users\sreev\javawebapp>docker build -t mywebim .
[+] Building 26.1s (8/8) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from dockerfile                                                                  0.1s
 => => transferring dockerfile: 197B                                                                                  0.1s
 => [internal] load metadata for docker.io/library/tomcat:9.0                                                         1.9s
 => [internal] load .dockerignore                                                                                     0.0s
 => => transferring context: 2B                                                                                       0.0s
 => [1/3] FROM docker.io/library/tomcat:9.0@sha256:76335d729032477a03a8ab1f49ccb740141aeb7b47498c98d00f77e0b79c2  22.5s
 => => resolve docker.io/library/tomcat:9.0@sha256:76335d729032477a03a8ab1f49ccb740141aeb7b47498c98d00f77e0b79c2b  0.0s
 => => sha256:de9be28b95192c91c53b88e11c9949a475df4ba95e71ca9a2ec1dad33873891c 17.46MB / 17.46MB                      7.6s
 => => sha256:4a896d1d091ba323ddf50ae3a4377aa973f595713bfeb90881a389943a59db20 15.43MB / 15.43MB                      7.6s
 => => sha256:cc1e0a39126833701e554d624057517d5d5863a8905ba55396be02741c872e51 2.28kB / 2.28kB                        1.6s
 => => sha256:beff1cfac18f9bcfd9e4903192bb73fc787e4e4b39163071b6d4d566aa87cb1e 140B / 140B                            1.6s
 => => sha256:c318c44e952a3391b86e5db6a972878c9a4f3f9b51cfbf6d22dcb6d19db6084f 92.39MB / 92.39MB                     17.8s
 => => extracting sha256:de9be28b95192c91c53b88e11c9949a475df4ba95e71ca9a2ec1dad33873891c                             0.8s
 => => extracting sha256:c318c44e952a3391b86e5db6a972878c9a4f3f9b51cfbf6d22dcb6d19db6084f                             2.1s
 => => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1                             0.0s
 => => extracting sha256:cc1e0a39126833701e554d624057517d5d5863a8905ba55396be02741c872e51                             0.0s
 => => extracting sha256:beff1cfac18f9bcfd9e4903192bb73fc787e4e4b39163071b6d4d566aa87cb1e                             0.0s
 => => extracting sha256:4a896d1d091ba323ddf50ae3a4377aa973f595713bfeb90881a389943a59db20                             0.4s
 => [internal] load build context                                                                                     0.2s
 => => transferring context: 370.22kB                                                                                 0.1s
 => [2/3] RUN rm -rf /usr/local/tomcat/webapps/*                                                                      0.7s
 => [3/3] COPY target/javawebapp.war /usr/local/tomcat/webapps/ROOT.war                                              0.1s
 => exporting to image                                                                                                0.5s
 => => exporting layers                                                                                               0.2s
 => => exporting manifest sha256:5784a403fb9a445bec216f9e1294d457f5f467b0c067d1ab39d34451755f9d1a                     0.0s
 => => exporting config sha256:432f848de59e012457c3b0cfc0f7c628602a5dd1db12ab0503f9c08cac839d15                       0.0s
 => => exporting attestation manifest sha256:a94b74e07f7b11bd746ccc2cde1f3f5c73fbaf2b0f446ba3f17a5714e7a8fbc1         0.0s
```

```
Command Prompt

 => [1/3] FROM docker.io/library/tomcat:9.0@sha256:76335d729032477a03a8ab1f49ccb740141aeb7b47498c98d00f77e0b79c2  22.5s
 => => resolve docker.io/library/tomcat:9.0@sha256:76335d729032477a03a8ab1f49ccb740141aeb7b47498c98d00f77e0b79c2b  0.0s
 => => sha256:de9be28b95192c91c53b88e11c9949a475df4ba95e71ca9a2ec1dad33873891c 17.46MB / 17.46MB                      7.6s
 => => sha256:4a896d1d091ba323ddf50ae3a4377aa973f595713bfeb90881a389943a59db20 15.43MB / 15.43MB                      7.6s
 => => sha256:cc1e0a39126833701e554d624057517d5d5863a8905ba55396be02741c872e51 2.28kB / 2.28kB                        1.6s
 => => sha256:beff1cfac18f9bcfd9e4903192bb73fc787e4e4b39163071b6d4d566aa87cb1e 140B / 140B                            1.6s
 => => sha256:c318c44e952a3391b86e5db6a972878c9a4f3f9b51cfbf6d22dcb6d19db6084f 92.39MB / 92.39MB                     17.8s
 => => extracting sha256:de9be28b95192c91c53b88e11c9949a475df4ba95e71ca9a2ec1dad33873891c                             0.8s
 => => extracting sha256:c318c44e952a3391b86e5db6a972878c9a4f3f9b51cfbf6d22dcb6d19db6084f                             2.1s
 => => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1                             0.0s
 => => extracting sha256:cc1e0a39126833701e554d624057517d5d5863a8905ba55396be02741c872e51                             0.0s
 => => extracting sha256:beff1cfac18f9bcfd9e4903192bb73fc787e4e4b39163071b6d4d566aa87cb1e                             0.0s
 => => extracting sha256:4a896d1d091ba323ddf50ae3a4377aa973f595713bfeb90881a389943a59db20                             0.4s
 => [internal] load build context                                                                                     0.2s
 => => transferring context: 370.22kB                                                                                 0.1s
 => [2/3] RUN rm -rf /usr/local/tomcat/webapps/*                                                                      0.7s
 => [3/3] COPY target/javawebapp.war /usr/local/tomcat/webapps/ROOT.war                                              0.1s
 => exporting to image                                                                                                0.5s
 => => exporting layers                                                                                               0.2s
 => => exporting manifest sha256:5784a403fb9a445bec216f9e1294d457f5f467b0c067d1ab39d34451755f9d1a                     0.0s
 => => exporting config sha256:432f848de59e012457c3b0cfc0f7c628602a5dd1db12ab0503f9c08cac839d15                       0.0s
 => => exporting attestation manifest sha256:a94b74e07f7b11bd746ccc2cde1f3f5c73fbaf2b0f446ba3f17a5714e7a8fbc1         0.0s
 => => exporting manifest list sha256:d888360b4f62120b6860dae5411b4f60bff382657a397c4b0bb38c1df4efd926               0.0s
 => => naming to docker.io/library/mywebim:latest                                                                     0.0s
 => => unpacking to docker.io/library/mywebim:latest                                                                  0.1s

C:\Users\sreev\javawebapp>docker run -d -p 8080:8080 --name javacontpractice mywebim
90bf4c5333fff901de9cbee48237b3ef997e10ce73da5d443e0e79dedd55aee6

C:\Users\sreev\javawebapp>
```

localhost:8080

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Practical on WAR with Maven,docker and Tomcat

```
C:\Users\sreev\javawebapp>docker stop javacontpractice
javacontpractice

C:\Users\sreev\javawebapp>type dockerfile
FROM tomcat:9.0
RUN rm -rf /usr/local/tomcat/webapps/*
COPY target/javawebapp.war /usr/local/tomcat/webapps/ROOT.war
EXPOSE 8080
CMD ["catalina.sh","run"]
C:\Users\sreev\javawebapp>
```